D07 · Warehouse KPIs, Metrics & Governance

04 · Warehouse S&OP; integration & capacity governance

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Warehouse S&OP integration brings the warehouse's capacity, throughput, cost and performance into the monthly sales and operations planning cycle, so capacity gaps are seen and resolved before the volume arrives.

2 WHY IT MATTERS

Demand plans are usually expressed in revenue and units, rarely in pallets, lines or labor hours. **If nobody translates the plan into warehouse throughput, the capacity constraint is discovered at the peak, when only expensive emergency solutions are left.** Additional shifts, hiring and investments are business decisions that belong in the S&OP forum.

3 HOW IT IS DONE

1. **Translate the demand plan into warehouse load.** Pallets, lines, orders and labor hours by month and site.
2. **Compare with available capacity.** Space, throughput and labor, with the gap quantified.
3. **Bring options to the supply review.** Shifts, temporary space, outsourcing or investment, with cost and lead time.
4. **Report performance and cost variance monthly.** Against the plan, with root causes.

IN PRACTICE

The warehouse must be an active S&OP participant — not a passive volume plan receiver

A warehouse that only receives the volume plan without participating in the S&OP cannot present its capacity constraints with sufficient advance notice.

The warehouse throughput forecast must be part of the S&OP plan

The S&OP projects order volume and revenue — but rarely projects warehouse throughput. The warehouse needs that forecast with sufficient advance to plan capacity.

Warehouse capacity investments must be approved in the S&OP — not in the warehouse

A decision to add an additional shift or hire 15 additional operators has budget implications requiring management approval in the correct forum.

The monthly S&OP cycle must include warehouse cost variance vs. budget

Actual cost per order vs. budgeted is a warehouse efficiency indicator management must see monthly.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Warehouse S&OP integration — first time the warehouse participates actively — The planning team makes demand forecasts without considering warehouse capacity.

Gap identified	Impact without integration	Result with integration
Q4 forecast projects 145% of current warehouse maximum throughput	HIGH	Without integration: gap discovered in Q3 — too late to act
E-commerce channel expansion requires VAS the current warehouse doesn't have	MEDIUM	No S&OP visibility: channel launched without the necessary VAS capacity

Gap identified	Impact without integration	Result with integration
Revenue growth plan (+22%) implies +35% throughput	MEDIUM	Without integration: capacity investments not planned in advance

Result: Post-integration: The warehouse will present its available capacity and capacity gaps in the monthly S&OP supply review. The first review revealed the Q4 gap 6 months in advance — sufficient time to implement an additional shift.

HOW IT IS MEASURED

Capacity Gap Detection Lead Time (months of advance notice for gap detection in S&OP)

Capacity Gap Detection Lead TimeMonths of advance from gap detection to projected gap date

Benchmark: >6 months of advance to implement solutions without urgency A capacity gap detected with <2 months of advance has no medium-term solution — only costly emergency solutions.

Warehouse Performance vs. S&OP Target %

Warehouse Performance vs. S&OP Target %(Actual warehouse KPIs / S&OP plan target KPIs) × 100

Benchmark: >95% of warehouse KPIs within the S&OP target range Consistently below-plan performance indicates the S&OP capacity assumptions are incorrect or the warehouse has execution problems not reflected in the plan.

WHAT YOU WOULD USE

S&OP Integration

SAP IBP — Integrated Business Planning — Module: Warehouse Capacity in S&OP + IBP FinancialSAP IBP module that integrates warehouse capacity plan with the demand plan and S&OP financial plan.

Power BI — S&OP Warehouse Dashboard — Module: Warehouse Capacity + Performance in S&OPPower BI dashboard for monthly warehouse S&OP review: available vs. projected capacity, performance KPIs, and cost variance vs. budget.

4 SO WHAT

Put warehouse capacity on the S&OP agenda. Translate demand into throughput, quantify gaps months ahead, bring costed options and decide them in the right forum.

From the SCRA v2.2 reference architecture · mrsupplychain.ai · © 2026 Ismael López · Mr. Supply Chain® México